🧅 Watch Cells Divide

Name: Class: Date:

Activity 1 · Hands-on · 🧪 In the lab · ⏱ 1 hour

🧅 Onion Root Tip Squash: See Mitosis

You will learn: Root tips divide actively, so a stained squash of a root tip shows every stage of mitosis.

YOU NEED

- An onion kept with its base in water for 3–4 days (roots 2–3 cm long)
- Acetocarmine stain
- 1N hydrochloric acid, a spirit lamp, a watch glass
- Slides, coverslips, blotting paper, forceps and a blade
- A compound microscope

⚠️ **Safety:** Hydrochloric acid is corrosive: handle it with the teacher and wash your hands. Warm gently and never let the acid boil dry.

STEPS

1. Cut 1 cm long root tips in the morning, when most cells are dividing.
2. Warm the tips gently in acetocarmine with 2–3 drops of 1N HCl for about 5 minutes.
3. Place one tip on a slide in a drop of stain and cut off the white end; keep only the dark tip.
4. Put a coverslip on it, cover with blotting paper and press gently with your thumb to squash it.
5. Look under low power, then high power, and find cells in each stage.

OBSERVATION TABLE

Stage	Number of cells seen	What you saw

QUESTIONS

1. Which stage did you see most often? Why?

2. In which stage could you count the chromosomes most easily?

3. Why are root tips chosen for this experiment?

WHAT IS HAPPENING? — THE SCIENCE

Root tips have a meristem whose cells divide by mitosis again and again. Acid softens the tissue so the cells spread out, and acetocarmine stains the chromosomes red. Most cells are in interphase because it takes more than 95% of the cell cycle; metaphase shows the chromosomes most clearly on the metaphase plate.

Key terms: mitosis interphase metaphase anaphase meristem

Activity 2 · Hands-on · 🏠 At home · ⏱ 45 minutes

📿 Bead Chromosomes: Model Mitosis and Meiosis

> **You will learn:** Moving model chromosomes through each stage shows why mitosis keeps the chromosome number and meiosis halves it.

YOU NEED

- Pop-beads or modelling clay in two colours (red from the mother, blue from the father)
- Paper clips to act as centromeres
- A chart paper with a big circle drawn on it (the cell)
- Thread for spindle fibres

> ⚠️ **Safety:** Keep small beads away from young children.

STEPS

1. Make one long and one short chromosome in each colour, so the cell has 2n = 4.
2. S phase: make an identical copy of each chromosome and join the two strands with a clip.
3. Mitosis: line all four up in the middle, then pull the sister strands to opposite sides. Count the chromosomes on each side.
4. Meiosis I: pair each red chromosome with its blue partner, swap a small piece (crossing over), then pull the partners apart.
5. Meiosis II: in each half, pull the sister strands apart again. Count the chromosomes in all four cells.

OBSERVATION TABLE

Step	Number of cells	Chromosomes per cell

QUESTIONS

1. How many cells and chromosomes did you get after mitosis? After meiosis?

2. What did swapping the beads show?

3. Why must meiosis happen before gametes are formed?

WHAT IS HAPPENING? — THE SCIENCE

In mitosis sister chromatids separate, so each daughter cell gets the full set (2n). In meiosis I the homologous partners separate, halving the number to n; meiosis II then separates the sisters. The swapped bead pieces show crossing over, which mixes the parents' genes and creates variation.

Key terms: **sister chromatids** **homologous chromosomes** **crossing over** **meiosis**

Digital activity

💻 Which Stage Am I?

> Tap an event, then tap the division in which it happens.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu1-ch10/2**

INTERPHASE · MITOSIS · MEIOSIS I · MEIOSIS II

Interphase	Mitosis	Meiosis I	Meiosis II

DNA replication (2C → 4C) · Centriole duplicates · Proteins for mitosis are made (G_2) · Chromosomes of a 2n cell on the metaphase plate · Sister chromatids of a 2n cell go to opposite poles · Two diploid daughter nuclei form · Synapsis of homologous chromosomes · Crossing over in pachytene · Homologous chromosomes separate · A dyad of cells forms · Centromeres split in a haploid cell · A tetrad of four haploid cells forms

> **You will learn:** Interphase prepares the cell; mitosis separates sister chromatids and keeps the number 2n; meiosis I separates homologues (2n → n) and meiosis II separates sister chromatids (n → n).

THINK & WRITE

1. Which events happen only in meiosis I?

2. Why is meiosis II called equational?